

# SAMPLE QUESTIONs (DSA)

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## UNIT-1 (Computer Basics, Algorithms, Flowcharts, Programs, C++ Basics)

1. Define the following terms with examples: (a) Compiler (b) Interpreter.
2. Differentiate between Primary Storage and Secondary Storage with suitable examples.
3. Explain the role of the Operating System as a resource manager.
4. Distinguish between Input devices and Output devices with two examples each.
5. Draw a neat flowchart to calculate the factorial of a given number.
6. Write an algorithm to find the largest of three numbers.
7. Explain the difference between Algorithm and Flowchart. Which one is easier for beginners to use? Why?
8. Explain the difference between Source Code and Object Code with a simple example in C++.
9. Write the structure of a basic C++ program with an explanation of each part.
10. What is the purpose of the main() function in C++? Why is it necessary?
11. List and explain the importance of header files in C++. Give two examples.
12. Identify and correct the logical error in the following C++ snippet:

```
int a = 5, b = 0;  
int c = a / b;  
cout << c;
```

14. Explain the difference between Syntax Error and Logical Error with examples.
15. Explain the use of `cin` and `cout` in C++ with an example program.
16. What is the significance of comments in a program? Demonstrate with a short example.
17. Differentiate between Compiler Linker and Loader.
18. Write an algorithm and draw a flowchart to calculate the sum of the first N natural numbers.
19. Convert the following algorithm into a C++ program:
- Algorithm:*
1. Input radius
  2. Compute  $\text{Area} = \pi * r * r$
  3. Print Area.
20. Explain the sequence of steps from writing a program to executing it on a computer system.
21. Draw a flowchart to check whether a given number is prime or not.
22. Write an algorithm to generate the Fibonacci series up to N terms.
23. Write a C++ program that reads marks of three subjects and prints the average and grade (A/B/C).
24. Differentiate between High-level language and Assembly language with suitable examples.
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## UNIT-2 (Tokens, Variables, Data Types, Operators, Expressions)

26. Define Tokens in C++. List different types with one example each.
27. Differentiate between Keywords and Identifiers in C++ with examples.
28. Write valid and invalid examples for variable naming conventions in C++.
29. Explain variable scope and lifetime with an example program.
30. Classify data types in C++ into Basic, Derived, and User-defined. Give one example of each.
31. Write a C++ program that demonstrates typecasting between float and int.
32. State the storage size (in bytes) of the following C++ data types: int, float, double, char.
33. Evaluate the following expressions (show steps):
  - (a)  $10 + 20 * 3$
  - (b)  $(100 / 5) + (8 \% 3)$
34. Explain operator precedence and associativity with an example expression.
35. Differentiate between Relational and Logical Operators with an example each.
36. Write a C++ expression using a conditional operator to find the maximum of two numbers.
37. Explain Bitwise AND (&), OR (|), and XOR (^) with the help of binary examples.
38. Evaluate step by step:

**int a = 5, b = 3, c;**

**c = a++ + ++b;**

39. Explain the difference between Arithmetic Operators and Assignment Operators with examples.

40. Evaluate the following and show your work:

(a)  $7 > 5 \ \&\& \ 10 < 15$

(b)  $!(4 == 4 \ || \ 3 > 5)$

41. Explain the difference between pre-increment ( $++x$ ) and post-increment ( $x++$ ) with a short code example.

42. Write a C++ program that reads two numbers and prints their sum, difference, product, quotient and remainder.

43. Demonstrate the use of `math.h` (`cmath`) functions: `sqrt()`, `pow()`, and `abs()` with small examples.

44. Write a C++ program to evaluate and display the result of the expression:

$(x + y) * z - (x / y) + (z \% y)$

for  $x = 10, y = 3, z = 5$ .

46. Write a program to swap two numbers using arithmetic operators (without using a third variable).

47. Trace the output of the following program:

`int x = 2, y = 3;`

`x = x++ * ++y;`

`cout << x << " " << y;`

48. Write a program to find the roots of a quadratic equation using the formula:

$\text{root1} = (-b + \sqrt{b*b - 4*a*c}) / (2*a)$

$\text{root2} = (-b - \sqrt{b*b - 4*a*c}) / (2*a)$

49. Explain the difference between Implicit and Explicit Typecasting with a program.

50. Given:

```
int a = 12, b = 5;
```

```
float c;
```

```
c = (a / b) + (float)a / b;
```

```
cout << c;
```

Predict the output and explain step by step.

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## Precedence & Associativity Numericals

51. Evaluate step by step:

```
int x = 10, y = 5;
```

```
int z = x + y * 2;
```

52. Predict the output:

```
int a = 8, b = 4;
```

```
cout << a - b + a * b;
```

53. Evaluate:

```
int a = 20, b = 4, c = 2;
```

```
int result = a / b * c;
```

54. Stepwise evaluation:

```
int x = 2, y = 3, z = 4;
```

```
int res = x + y * z - y;
```

55. Find output:

```
int a = 5, b = 2;
```

```
cout << a % b * a + b;
```

**56. Trace:**

```
int a = 5, b = 3;
```

```
int result = a++ + ++b * 2;
```

**57. Predict output:**

```
int x = 7, y = 2;
```

```
cout << x > y && x < 10 || y == 0;
```

**58. Evaluate:**

```
int a = 4, b = 6;
```

```
int result = a & b | a ^ b;
```

**59. Output?**

```
int a = 2, b = 3, c = 4;
```

```
int result = a + b * c / a % b;
```

**60. Trace:**

```
int x = 10, y = 5;
```

```
int result = (x > y) ? x + y : x - y;
```

**61. Stepwise evaluation:**

```
int a = 5, b = 10, c = 2;
```

```
int res = a + b > c && b - a < c;
```

**62. Complex evaluation:**

```
int a = 3, b = 4, c = 5;
```

```
int result = a++ * --b + c / a;
```

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